8 August 1974
SUPERSEDING
MIL-S-45527(0rd)
9 April 1962

MILITARY SPECIFICATION

SPRINGS, STRANDED WIRE, HELICAL COMPRESSION

(FOR SMALL ARMS APPLICATIONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers stranded wire helical compression springs formed of three or seven round wires twisted together.

1.2 Classification.

1.2.1 Types: Stranded wire springs shall be of the following types, as specified (see 6.l and 6.2).

Type I - Wire, Steel, Music
Type II - Wire, Steel, Corrosion-Resisting

1.2.2 Classes. Stranded wire s rings shall be of the following classes, as specified (see 6.1 and 6.2).

Class 1 - Three-Wire Strand
Class 2 - Seven-Wire Strand

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue on date of invitation for bids, form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-W-13855 - Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-14232 - Parts, Equipment and Tools for Army Material, Packaging and Packing of.
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of.

STANDARDS

Military - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-105 - Quality Assurance Terms and Definitions.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

* 3.2 Materials, construction and design. Springs shall conform to the materials, construction, and design requirements specified herein) on the applicable drawings (see 6.1 and 6.2) and in MIL-W-13855.

* 3.2.1 Strain relief. Unless otherwise specified (see 6.2), cold-wound springs just formed shall be given a suitable strain (stress) relief treatment conforming to 3.2.1.1 or 3.2.1.2.

* 3.2.1.1 Carbon steel springs. Bake for 30 minutes at 525° ± 50°F.

* 3.2.1.2 Corrosion-resisting steel springs. Bake for 30 minutes at 800° ± 50°F.

3.2.2 Direction of helix. The direction of helix shall be as specified on the applicable drawing. The helix of the spring coils shall be opposite in direction to the helix of the wires in the strand; e.g., a right-hand coiled spring shall be formed from a left-hand helix stranded wire.

3.2.3 Three-wire strand. The pitch, or lay, of the strands shall be between 5 and 5 1/2 times the strand diameter.

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3.2.4 Seven-wire strand. Seven-wire strand shall be composed of six outer wires wound around a common core wire. The diameter of the core wire shall be 5 to 15 percent larger than the outer wire diameter. Pitch for seven-wire strands shall be between 5 and 5 1/2 times the strand diameter.

3.2.5 Construction of ends. End construction shall be as specified on the applicable drawing. Brazing or welding material shall not interfere with the operation of the spring. After brazing or welding, residual flux shall be neutralized and removed by an appropriate solvent to prevent subsequent corrosion.

* 3.2.6 Pitch. The pitch of the active coils shall be as specified on the applicable drawing. Unless otherwise specified, end coils of compression springs with square and ground ends shall decrease uniformly from full pitch to a pitch of one wire diameter.

* 3.2.6.1 Uniformity. The pitch of the active coils of springs shall be sufficiently uniform so that when compressed, unsupported laterally, to a length equal to 80 percent of deflection, see Figure 1, none of the active coils shall be in contact. When the design of the spring is such that it cannot be compressed without support, this requirement shall be waived.

3.2.7 Solid height. The solid height shall be as specified on the applicable drawing. The solid height of a helical compression spring shall be the perpendicular distance between the plates of a government approved testing machine, when the spring is compressed with a test load to bring all coils in contact. In no case shall this test load exceed by more than 50% the load beyond which no appreciable deflection takes place. nor shall the test load be held for any appreciable length of time.

* 3.3 Performance. The springs shall contain the required load-height relationships specified on the applicable drawing. Unless otherwise specified, springs shall be preset prior to performance testing. Testing shall be as specified in 4.5.3.2.

* 3.4 Marking. Springs shall be marked in accordance with MIL-W-13855.

+ 3.5 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

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* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

* 4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

* 4.4 First article inspection. The first article (initial production unit(s))shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

* 4.5.1 Inspection lot. Unless otherwise specified, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855.

* 4.5.2 Examination of springs shall be as specified herein and in MIL-W-13855. The examination provisions should be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of chance in the characteristic by subsequent operat ions. Reinspection of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling and that adequate inspection records are maintained.

* 4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQL’s) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQL's apply to each listed characteristic.

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b. Where "Visual" is specified as the inspection method for protective coating, the coating shall be visually examined for completeness, uniformity in appearance, scratches, and worn or bare spots.
c. Where "Visual" is specified as the method for dimensional or other general drawing characteristics, the characteristic shall be either scaled or compared with a specimen of known acceptable quality that has been established as an inspection standard, for conformance to the requirements of the drawing.
d. Where "SMTE" (Standard Measuring and Test Equipment) is specified as the method of inspection the contractor may use any type of industry-developed, commercially available, multi-usage equipment or special inspection and/or testing equipment approved by the Government,

* 4.5.2.1.1 Springs. Springs shall be examined in accordance with the classification of defects, as applicable, In Table I.

Table I

Class	Characteristics	AQL	Inspection Method
Critical:	None defined		
Major:			
101	Outside diameter - incorrect (3.2)	0.65	SMTE
102	Inside diameter - incorrect (3.2)	0.65	SMTE
103	Free length - not as specified (3.2)	0.65	SMTE
104	Solid height - incorrect (3.2.7)	0.65	SMTE
105	Direction of helix - not as specified (3.2.2)	0.65	Visual
106	Pitch of strands - incorrect (3.2.3) (3.2.4)	0.65	SMTE
107	End Construction - incorrect (3.2.5)	0.65	SMTE
108	Pitch of coils - incorrect or not uniform (3.2.6) (3.2.6.3.)	0.65	SMTE
109	protective Finish - missing or defective (3.2)	0.65	Visual
Minor:			
201	Marking - incorrect or missing (3.4)	2.5	Visual
202	workmanship (3.5)	4.0	Visual

4.5.3 Testing.

* 4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on complete springs. If test requirements cited herein are not met, acceptance of the springs shall be deferred and the contractor shall accomplish as applicable, the following actions:

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a. Conduct a failure analysis study performing a dimensional) physical and visual examination of the springs.
b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine springs to insure that materiel containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review end approval prior to submitting a reconditioned lot or reconditioned springs for retest.

4.5.3.2 Performance testing. The contractor shall test each stranded wire spring for the performance requirement (see 3.3) using the test method specified in 4.6.1 until a process average of one percent defective or less is attained for 200 consecutive stranded wire springs. When a process average of one percent defective or less is attained for 200 consecutive stranded wire springs, the contractor shall test a sample taken from each inspection lot in accordance with MIL-STD-105. Acceptance shall be based on an AQL of 0.65 percent defective. If an inspection lot is rejected, 100 percent testing shell be reinstated until a process average of one percent defective or less is again attained for 200 consecutive stranded wire springs, at which time the contractor shell return to the above sampling plan.

* 4.5.3.3 Material and processes. Materials and processes shall be tested in accordance with the applicable specifications or requirements referenced on the applicable drawing and this specification. When the contractor's certification of chemical analysis or certification of tests is available, they will be accepted for the following:

* 4.5.3.3.1 Materials. Certification that the materials used, conform to the requirements of 3.2.

* 4.5.3.3.2 Heat treatment. Certification that the heat treatment used conforms to the applicable drawing.

* 4,5.3.3.3 Strain relief. Certification that the strain relief methods used conform to requirements of 3.2.1 through 3.2.1.2 and the applicable drawing.

* 4.5.3.4 Packaging, examination and testing. Unless otherwise specified (see 6.2). the packaging examination and testing shall be in accordance with MIL-P-14232.

4.5.4 Inspection equipment.

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* 4.5.4.1 Acquisition, calibraton, maintenance and disposition. Unless otherwise specified (see 6.2), responsibility for acquisition, calibration, maintenance and disposition of acceptance inspection and test equipment shall be in accordance with MIL-I-45607.

* 4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

* 4.6.1 Performance test. Springs shall be tested using a Government approved testing machine. The springs shall be compressed to the maximum test height and the load shall be measured and recorded. The springs shall then be compressed to the minimum test height and the load shall be measured and recorded. All of the above measurements shall be made during the compression stroke. Precautions shall be taken to assure that there Is no fictional bind during the compression stroke. Unless otherwise specified on the applicable drawing, the springs shall be compressed a minimum of five times from the free height to the solid height before performing the performance test.

5. PREPARATION FOR DELIVERY

* 5.1 Pilot Pack. The pilot pack shall consist of a spring or springs as required by and packaged in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2 and 6.3) packed level C and forwarded in accordance with 3.1.

* 5.2 Preservation, packaging, packing and marking. Springs shall be preserved, packaged, packed and marked in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2 and 6.3).

6. NOTES

* 6.1. Intended use. This specification is intended to cover, but not be limited to, procurement of the following springs:

Drawing	Nomenclature
C7191544	Spring, Driving (For M37 Machine Gun).
B7269087	Spring, Firing Pin (For M60 Machine Gun).
B7269303	Spring, Driving (For M60 Machine Gun).
B7791254	Spring, Helical Compression (For M85 Machine Gun).
C8448799	Spring, Helical (For M219 Machine Gun).

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6.2 Ordering data. Procurement documents should specify:

Title, number, and date of this specification.
b. Type and class required (see 1.2.1 and 1.2.2).
c. List of drawings and specifications pertinent to the spring(s) showing applicable revision dates.
d. Responsibilities for acquisition, calibration, maintenance and disposition of acceptance inspection equipment, if different (see 4.5.4.1).
e. Requirements for submission of first article (see 3.1, 4.4 and 5.1).
f. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2).
g. Packaging, examination end testing, if different (see 4.5.3. h).
h. Strain relief, if different (see 3.2.1).

* 6.3 The following Packaging Data Sheets are applicable to the springs covered by this specification as specified in the contract (see 6.2).

Packaging Data Sheets:

P7191544 - Spring, Driving.
P7269087 - Spring, Firing Pin.
P7269303 - Spring, Driving.
P7791254 - Spring, Helical Compression.
P8448T99 - Spring, Helical.

* 6.4 Management control system requirements. When warranted, the contract should specify the application of MIL-I-45208 on the Management Control Systems Summary List, DD Form 166o.

* 6.5 Unless otherwise specified (see 6.2 d) the contract should specify the application of MIL-I-4560T and MIL-C-45662 on the Management Control System Summary List, DD Form 1660.

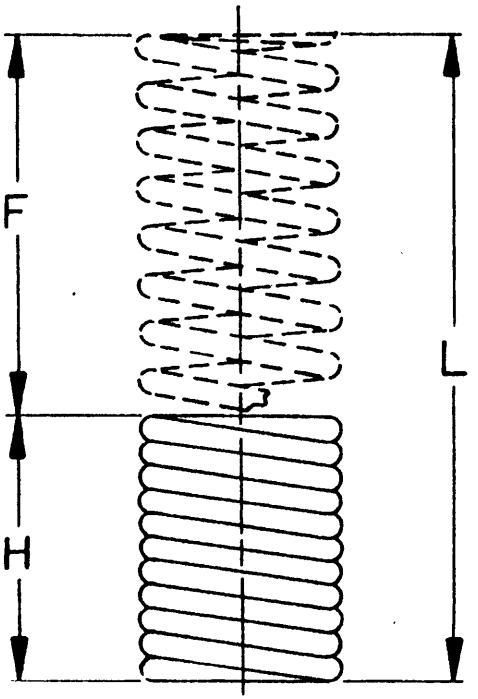
6.6 The margins of this specification are marked with an asterisk to Indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC

Preparing Activity:
Army-WC

Project Number:
1005-A516

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TEST DIMENSIONS

L = FREE LENGTH
H = SOLID HEIGHT
F = TOTAL DEFLECTION

SPRING CHARACTERISTICS

FIGURE 1

MIL-S-45527A(WC)

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DOCUMENT IDENTIFIER AND TITLE

MIL-S-45527A(WC); Springs, Stranded Wire, Helical Compression

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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MILITARY SPECIFICATION

SPRINGS, STRANDED WIRE, HELICAL COMPRESSION
(FOR SMALL ARMS APPLICATIONS)

This notice should be filed in front of MIL-S-45527A(AR), dated 8 August 1974.

MIL-S-45527A(AR) has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Army-AR

AMSC N/A
FSC 1005
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